

NeuroGuard - A Smart Wearable AI-Driven Device for Early Stage Detection and Progressive Assistance in Alzheimer's Disease Patients

**Project report in partial fulfillment of the requirement for the award of the degree of
Bachelor of Technology
in
Computer Science and Engineering (Artificial Intelligence & Machine Learning)**

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This is to certify that the project titled **NeuroGuard - A Smart Wearable AI-Driven Device for Early Stage Detection and Progressive Assistance in Alzheimer's Disease Patients** submitted by **Prantika Ghosh**(University Registration No. 304202300901073 of 2023 - 2024), **Pijush Banerjee**(University Registration No. 304202300901069 of 2023 - 2024), **Rishita Paul**(University Registration No. 304202300901090 of 2023 - 2024), **Priyanshu Ghosh**(University Registration No. 304202300901077 of 2023 - 2024), and **Veerat Pandey**(University Registration No. 304202300901027 of 2023 - 2024), students of the Institute of Engineering & Management, Kolkata, school of University of Engineering and Management, Kolkata, in partial fulfillment of requirement for the degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning), is a bonafide work carried out by them under the supervision and guidance of Prof. (Dr.) Sudipta Sahana during 6th Semester of academic session of 2026 - 2027. The content of this report has not been submitted to any other university or institute. I am glad to inform that the work is entirely original and its performance is found to be quite satisfactory.

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TABLE OF CONTENTS

ABSTRACT	8
CHAPTER – 1: INTRODUCTION	9
CHAPTER – 2: LITERATURE SURVEY	
2.1 Overview of Alzheimer’s Disease and Importance of Early Detection	10
2.2 Traditional Diagnostic Techniques	10
2.3 Appearance of Artificial Intelligence in Alzheimer’s Research	11
2.4 Machine Learning and Deep Learning Approaches for Detection	11
2.5 Speech and Language Analysis as Cognitive Biomarkers	11
2.6 Wearable Technology for Alzheimer’s Monitoring	12
2.7 Multimodal Data Integration and Predictive Modeling	12
2.8 Conversational Assistive Technology for Alzheimer’s	12
2.9 Limitations of Existing Studies and Research Gaps	13
2.10 Early Diagnostic Methods	13
2.11 Artificial Intelligence in Alzheimer’s Detection	13
2.12 AI in Speech and Behavioral Biomarkers	14
2.13 Wearable Technologies in Alzheimer’s Care	14
2.14 Limitations and Data Challenges	14
2.15 Integration of AI and Wearable Systems	15
2.16 Real-Time Processing and Edge AI	15
2.17 Assistive Voice Technology in Cognitive Care	15
2.18 Summary of Research Gaps	16

CHAPTER – 3: PROBLEM STATEMENT

3.1 Background and Context 16

3.2 Challenges in Current Alzheimer’s Management 17

3.2.1 Monitoring Challenges 17

3.2.2 Support and Safety Issues 17

3.3 Identified Research Gaps 18

3.4 Statement of the Core Problem 18

3.5 Need for the Proposed System 19

3.6 Scope and Expected Impact 19

CHAPTER – 4: PROPOSED SOLUTION

4.1 Introduction and Overview 20

4.1.1 Core Functionalities 20

4.1.2 Objectives of the Proposed Solution 20

4.1.3 Target Users 21

4.2 System Architecture 21

4.2.1 High-Level Interaction Flow 21

4.3 NeuroGuard Sensor Unit (Wearable Device) 22

4.3.1 Form Factor 22

4.3.2 Integrated Sensors 23

4.3.3 Processing and Communication 24

4.3.4 Power Management 24

4.3.5 User Interface Elements 24

4.4 NeuroGuard Companion App (Mobile Application) 25

4.4.1 Role and Platform 25

4.4.2 Data Synchronization 25

4.4.3 Patient Interface (Simplified) 25

4.4.4 Caregiver Interface (Comprehensive Dashboard) 26

4.4.5 Offline Capabilities 26

4.5 NeuroGuard Cloud Platform	27
4.5.1 Purpose	27
4.5.2 Components	27
4.5.3 Security and Privacy	28
4.6 NeuroGuard AI Core	28
4.6.1 Early Detection Module	28
4.6.2 Model Architecture (Primary – LightGBM)	29
4.6.3 Progressive Assistance Module (Real-time/Near Real-time)	29
4.7 Data Flow and Interaction Scenarios	30
4.7.1 Detailed Data Journey	31
4.7.2 Key Interaction Scenarios	32
4.8 Implementation Considerations	33
4.8.1 Hardware Prototyping	33
4.8.2 Software Development	33
4.8.3 Data Acquisition & Ethics	34
4.8.4 Clinical Validation	34
4.8.5 Scalability and Deployment	34
4.9 Expected Outcomes and Benefits	35
4.9.1 For Patients	35
4.9.2 For Caregivers	35
4.9.3 For Clinicians	35
4.9.4 For Healthcare Systems Research	36

CHAPTER – 5: EXPERIMENTAL SETUP AND RESULT ANALYSIS

5.1 Introduction	36
5.2 Experimental Setup	36
5.2.1 Data Acquisition Protocol	36
5.2.2 Study Cohort	37
5.2.3 Data Preprocessing	38
5.2.4 Hardware and Software Environment	39

5.3 Feature Engineering and Extraction	39
5.3.1 EEG Features	39
5.3.2 Cardiovascular Features (PPG/ECG)	40
5.3.3 Electrodermal Activity (EDA) Features	40
5.3.4 Motion and Gait Features (IMU)	41
5.3.5 Other Features	42
5.3.6 Final Feature Vector	43
5.4 Machine Learning Model: LightGBM with SMOTE	44
5.4.1 Model Selection	44
5.4.2 Handling Class Imbalance: SMOTE	44
5.4.3 Hyperparameter Tuning: GridSearchCV	45
5.5 Result Analysis	45
5.5.1 Overall Model Performance	46
5.5.2 Performance Metrics	46

CHAPTER – 6: CONCLUSION & FUTURE SCOPE

6.1 Conclusion	48
6.2 Future Scope	49
6.2.1 Rigorous Clinical Validation and Real-World Data Acquisition	49
6.2.2 Wearable Hardware Advancement	49
6.2.3 AI Core Evolution	50
6.2.4 Pathway to Clinical Implementation and Importance	51
6.2.5 Long Term Vision	52

REFERENCES	53
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ABSTRACT

Alzheimer's disease is a growing health care problem affecting millions of people but traditional methods such as MRI or PET scans are extremely expensive and impossible for early continuous monitoring. This research proposes a new device, NeuroGuard, a wearable artificial intelligence device with an embedded voice assistant that will promote early detection and patient assistance. By looking for physiological and behavioral parameters such as heart rate, sleep patterns and speech patterns, NeuroGuard can utilize the Artificial Intelligence models to search for small changes, to detect the very early changes related to cognitive impairment in the patient's daily environment. The system operates in a two part fashion, with the voice assistant being a help to the patients, promoting medication compliance, safety and interesting dialogue to help with confusion and distract them from it, while acting as a data monitor for family monitoring of the patients progress and to alert family of any changes detected during the day through a web based dashboard. This marriage of monitoring and interactive helping will hopefully lessen the gap between patient independence and clinical observation and research. Initial prototypes have been developed to demonstrate the theoretical possibilities and future research will be directed toward clinical trials to show their validity in supporting the correct diagnoses that will eventually lead to advanced neurocognitive care through artificial intelligence that shows symptoms of early cognitive impairment. This will drastically change the lives of such patients.

Keywords: Alzheimer's disease, Cognitive, Monitoring, Prototype, Neurocognitive, impairment.

1. INTRODUCTION

A brain condition known as Alzheimer's disease gradually decreases memory, mental abilities, and the capacity for handling everyday activities. It is seen that those impacted will increase significantly owing to an expanding elderly demographic worldwide. Although significant progress has been made in understanding brain functions through different research and imaging techniques like fMRI. The onset of Alzheimer's progresses gradually without immediate signs for decades until recognizable cognitive changes appear. Modern diagnostics depend upon physical exams, cognitive assessments like the Montreal Cognitive Assessment, brain imaging techniques such as MRI and positron emission tomography, and spinal fluid analyses for checking purposes. Despite their precision, these techniques often come at high costs, require extensive invasions of privacy, and usually occur when substantial memory loss has already been observed. Consequently, the chance for timely checking and preventative measures is often overlooked.

It has become necessary to find non-invasive and continuous methods for monitoring people who are at risk of mental decline so that they can be detected early on before reaching a clinical stage. There are now many options for using the latest in intelligent wearables, IoT devices, and AI systems to provide an entirely new way to provide care in treating Alzheimer's disease. The application of these technologies, along with other forms of technology, will transform all of health care from a "hospitals only" model for diagnosing patients, to a "patient-centered" model for delivering constant, continuous, health care in the home by means of connected medical monitoring systems. NeuroGuard is a project which has for its object the invention of an intelligent medical monitoring apparatus with artificial intelligence and internet of things capabilities specifically designed for the detection of possible cognitive functions in patients suffering from Alzheimer's disease before they become of serious proportions, thus allowing the greatest individual attention to each degree of assistance to be present as the disease passes through its various stages. Despite the considerable progress made in both the early apprehension and cure of the disease highly important challenges still remain. Many patients are treated too late because their brains have been so severely damaged that a proper diagnosis can not readily be made.

Routine medical examinations, on the other hand, are infrequent and do not give an opportunity for continuous observations which would show important behavioral or health changes. Traditional diagnostic methods though sometimes reliable are involved with invasiveness, high costs and limited availability from the fact that diagnosis cannot be made readily in the unvisited or impoverished areas. Further than that, the errors in our present method of health care delivery are aggravated by the absence of much of the equipment for the performance of regular monitoring of mental health encouragement, mental peace, and safety applications. Attention to patients suffering from Alzheimer's disease is especially necessary for them because of the great risks of be able to lose their way, or of bumps, or of mental confusion which indicate the necessity of alert signalling devices for their immediate health in real time.

NeuroGuard's principal objective is the development of an intelligent health monitor for people who are vulnerable to a memory impairment of some sort, enabling them to realize symptoms of deterioration before it is too late and thus to present succession of treatment. The project's main objectives are to produce Early Detection, which is to develop an integrated unit, to gather physical measurements such as pulse rates, as well as mental measurements such as skin measurements which can be determined through skin conductance, to obtain the various measurements which are of a physical nature, such as the pulse rates. The next objective is Stage Prediction, that is, the determination of the degree to which a known disease is mild, medium, or severe, by known approaches presently in being in the medical sphere from past patient analysis, so that it can be medically serviced by physicians. The third purpose of the project is to obtain Continuous Surveillance and Service, through such means as instant notification and alarms when it has been discovered that a person has fallen, through establishing Zone G.P.S. located, thus providing reassurance, not only for the patient himself but also for the person looking after him, to appreciate the degree to which the patient is in a secure condition.

A fourth aim is Personalised Cognitive Assistance, the obtaining of which is only possible through a discovery of a flexible approach to the provision of reminders and ever-increasing personalisation of matter such as medication reminder notices, scheduled alert notices, reminders and psychological assistance, and nurse treatment of patients to assist in their processes of active involvement in the treatment of the individualized person. Finally, all operations will be dependent upon a secure Data Security and Privacy arrangement with respect to Cloud Computing, with strict confidentiality whilst permitting selective dissemination of information to approved people. NeuroGuard aims to produce an advanced, constant state of health treatment, leading to enhancement of quality of life with lower expenses incurred.

2. LITERATURE SURVEY

2.1. Overview of Alzheimer's Disease and Importance of Early Detection

Alzheimer, a neurodegenerative chronic disorder with predominant memory effects, behavior, ability to reasoning. The primary cause for dementia is likely AD, responsible for 60–70% cases globally. The condition is defined by Amyloid-beta plaque and tau protein tangles in the brain. It plays significant role in neuronal death and cognitive rejection. Gradually the symptoms arise and progress in severity from time to time, eventually it disrupts normal daily activities and independent living. More than 55 million people are globally impacted and the number can be triple by 2050 with life expectancy rising. Alzheimer touches more than just the patient in society. Families suffer physical, emotional, and financial pressures.[19] Proper monitoring must be done early and constantly to keep disease progression in control. Allow treatment in earlier stages, keep the quality of patient life intact. The diagnostic methods such as MRI and CSF are costly and not generally available in rural settings. Delayed diagnosis can cause extensive brain damage. So, scientists are now looking for low-cost and real time monitoring strategies using AI, wearables and wearables to identify early cognitive decline.[19]

2.2. Traditional Diagnostic Techniques

General Alzheimer's clinical diagnosis depends on neuroimaging, neuropsychological testing & biochemical examination. Structural changes are detected in MRI imaging in the brain like atrophy of the hippocampus, while metabolic activity is detected by PET imaging. CSF examination quantifies concentrations of amyloid-beta and tau proteins. They play an important role in Alzheimer's pathology.[20] Though these tools are too precise in clinical research. Although they possess several limitations. MRI and PET scans have specialized equipment, specialized staff & too expensive which limits their use to university research and major hospitals. CSF analysis includes lumbar puncture, which is an invasive and painful process non-appropriate for repeated observation. Cognitive testing such as MMSE or MoCA gives trusted information but this is subjective and can be biased or influenced by the patient's surroundings. These constraints demonstrate the reachable technologies' importance that can be available for scaling and have the potential to facilitate preliminary detection and long-term observation away from clinical setups.[20] Neuropsychological tests are established analysis and digital adaptations for preliminary cognitive declination detection. Our research started by conducting a comprehensive literature survey of prior investigations to identify gaps related to the detection of Alzheimer's disease and support provided to patients with Alzheimer's. The initial step of this process was critical to building a design and creating algorithms.

2.3. Appearance of Artificial Intelligence in Alzheimer's Research

The past ten years have witnessed immense expansion in the application of machine learning (ML) and AI to process biomedical and behavioral data for the early detection of Alzheimer's. AI algorithms are capable of handling immense amounts of multimodal data such as imaging, speech, motion, and physiological signals to identify minute patterns that are not detectable to human clinicians.[6] There have been early works involving feature extraction from MRI and PET scans for classifying patients as normal, mild cognitive impairment (MCI), or Alzheimer's type. Techniques of deep learning like convolutional neural networks (CNNs) have been employed in automatically learning spatial features of brain anatomy related to disease. Recurrent neural networks (RNNs) and long short-term memory (LSTM) networks have been employed on longitudinal data to forecast future cognitive decline. These AI-driven diagnostic tools have been shown to have good accuracy, but their dependence on imaging data obtained in a clinical setting hinders deployment in real-world settings. The current shift is toward fusing streams of wearable and behavioral data, which can be continuously and nonintrusively acquired.[6]

2.4. Machine Learning and Deep Learning Approaches for Detection

Machine learning classifiers like support vector machines (SVM), random forests, and k-nearest neighbors (KNN) have been used for Alzheimer's classification in the past with imaging and neuropsychological datasets. These models learn discriminative features such as gray matter volume, cortical thickness, and metabolic activity patterns. With advances in deep learning, feature engineering has been mostly automated and detection performance has

enhanced. CNNs are especially robust for MRI and PET image analysis because they can learn spatial hierarchies. Autoencoders and GANs have also been used for unsupervised feature learning and data augmentation, taking care of small dataset sizes. In addition, multimodal deep networks which integrate imaging with clinical or demographic information have been found to have higher prediction accuracy than unimodal models.[18] Nonetheless, steep computation requirements and privacy issues with data prevent deployment in home or wearable settings. Research focuses on lightweight models on-device and federated learning approaches to preserve privacy while supporting high inference efficiency.

2.5. Speech and Language Analysis as Cognitive Biomarkers

Speech and language have become effective, non-invasive biomarkers to identify early cognitive deterioration. Changes in vocabulary, syntax, fluency, and semantic coherence are present in Alzheimer's patients much before symptoms become clinically apparent. Research using datasets like DementiaBank has demonstrated that linguistic abnormalities could be identified by AI models using natural language processing (NLP) and acoustic feature extraction. Methods such as automatic speech recognition (ASR), prosodic analysis, and semantic embedding have been utilized to analyze features such as hesitation, repetition of words, and sentence complexity. Architectures like BERT and LSTM based networks show the metrics of accurate differentiation between Alzheimer's and normal subjects.[18] There is complementary information of mood swings which is provided by emotion and sentiment analysis of speech signals which commonly accompanies cognitive decline. The great advantage of speech diagnostics is their simplicity. Where speech analytics require only a simple appliance like microphone, similar things can be achieved with help of smart phone or voice enabled wearable devices. In real world check, the acoustic noise and language heterogeneity is on restricting side.[12]

2.6. Wearable Technology for Alzheimer's Monitoring

Continuous and real time monitoring systems of physiological and behavioral signals have transformed the technologies in the wearable arena.[2] Such devices can provide insight by having sensors, such as accelerometers, gyroscopes, heart-rate monitors, and skin conductance sensors, into a patient's activity level, quality of sleep, and stress level paid.[5] In research related to Alzheimer's disease, wearables have gathered data on movement abnormalities, clinical falls, and circadian rhythm abnormalities, which are all characteristic of cognitive decline.[2] Smartwatches and fitness trackers are also able to collect subtle changes in movement or fidgeting, which tend to track with disease worsening. When married to AI algorithms, predictive models can be created by these signals that can help in identifying early deterioration. Although they hold promising deliverables, recent wearable devices are not that great in care technologies which are disease-specific. Most of them are primarily designed for general monitoring of health and they do not prioritize integrating clinical data. Thus for Alzheimer's care as standalone devices they remain restricted. The idea of developing dedicated systems are influenced by this shortcoming for example NeuroGuard which is designed specifically for cognitive health purposes.

2.7. Multimodal Data Integration and Predictive Modeling

A single biomarker never can completely showcase Alzheimer's development and human cognition is multifaceted. Related to these diseases, human cognition is multifaceted. Predictive ability can be improved with these additions by alterations from structural, functional and behavioral perspective. Recent researches have focused significantly on integrating data from wearable sensors, smartphones, and clinical evaluations that can help in creating diagnostic frameworks that are more efficient.[14] Data fusion methods are employed by multimodal machine learning models namely early fusion (feature-level), late fusion (decision-level), and hybrid approaches to combine various data flows.[9] Independent single modality systems can be surpassed by these models and can learn in dealing with blank inputs or the inputs containing noise. Though multimodal systems face critical technical challenges related to synchronization, dimensionality of information and privacy guard. To address this complex nature at the same time as retaining real-time responsiveness, edge computing and cloud-based design are under consideration.[18]

2.8. Conversational assistive tech for Alzheimer's

Aiding a vital function is held by Assistive devices. Alzheimer's patients with activities of daily living, managing safety, and emotional status.[7] Patient autonomy is enhanced by advancements like GPS tracking, fall detection

alert and reminder systems and minimizing caregiver burden.[3] For health care use advanced voice assistants such as Apple Siri, Alexa have been modified that can set reminders for medicines and appointments can be scheduled also having basic conversational facilities. Although these assistants fail in personalization and medical awareness are not taken into consideration.[7] Disease-specific conversational AI is the primary focus point for new research that can engage in empathetic conversation, memory-support, and cognitive tasks. NeuroGuard develops on this concept by integrating an intelligent voice assistant within wearable sensors. To caregivers this not only offers reminders and emotional support but the assistant also communicates with the base AI foundation to seamlessly deliver timely warnings, safety alerts and real-time feedback This conversation-cognition synergy establishes a new path in person-focused Alzheimer's care.

2.9. Limitations of Existing Studies and Research Gaps

In AI-focused Alzheimer's research, while remarkable advances have been reported, there still exists various limitations. Small, homogeneous datasets are still the base for numerous researches that pose problems for generalizability to populations. Datasets related to imaging are way costly to obtain and tend to not be diverse in terms of age, ethnicity, and stage of disease. Whereas it is less costly to obtain Speech datasets, while it faces issues of language and accent variability. Research focused on wearables in its nascent stages.[12] The large number of devices are driven towards mobility and physical activity while it fails to address aspects which are cognitive or emotional. On top of that, one important consideration is data privacy and security especially in the time when health information is being conveyed via networks which are sensitive. Standard frameworks for data fusion, ethical data gathering, and verification in real-world settings are also not available. These gaps should be mitigated up by future research forming inclusive datasets, AI systems which are privacy enhanced and models that can be intercepted that clinicians and on which carers can rely on. Among engineers, ongoing joint work , medical practitioners, and psychologists will be essential for creating prototypes into tools which are relevant from clinical perspective.[12]

2.10. Early Diagnostic Methods

Primarily Alois Alzheimer described Alzheimer's disease in 1906 and was diagnosed by confirmation which were based solely on postmortem examination of the brain. Decades after decades diagnosis mostly be done considering clinical symptoms such as memory-impairment, confusion and changes in behaviour. Such symptom-based assessment was not very accurate, especially during early phases when therapeutic intervention would be most effective. At the turn of the 20th century, diagnostic capabilities involving neuroimaging (CT, MRI) and cerebrospinal fluid (CSF) analysis made in vivo visualization of brain abnormalities possible.[19] Important pathological features have been revealed by these methods such as amyloid-beta ($A\beta$) plaques and neurofibrillary tangles made of tau proteins. Although they have been successful, they are still costly, invasive, and hospital-based, which restricts their broader utilization.

2.11. Artificial Intelligence in Alzheimer's Detection

The combination of AI and deep learning has enormously improved the analysis of neuroimaging data. MRI and PET scans extensively utilize CNNs to analyze and classify Alzheimer's stages in automatic mode. Deep learning models were capable of over 90% accuracy in discriminating Alzheimer's patients from healthy controls based on multimodal imaging. Equally CNNs which integrate hybrid systems with RNNs have been utilized for temporal tracking of brain degeneration from time to time.[14] Feature extraction has been minimized by these computer-based approaches by hand and is able to identify subtle brain patterns that change undetectable to human experts.

2.12. AI in Speech and Behavioral Biomarkers

Besides imaging, AI played a remarkable role in examining speech, walk, and facial features as possible digital biomarkers.[13] Early cognitive impairment can be indicated Patterns in language e.g., vocabulary density, pause rate, and syntactic complexity. In 2019 Fraser et al showcased that machine learning algorithms trained on spontaneous speech can do prediction in cognitive impairment with good promises of sensitivity. Similarly, reduced mobility or atypical sleep patterns like shifts in behaviour have been recorded with the help of wearable sensors and verified with the help of AI classifiers, for example Random Forests and SVMs.[10, 11] Gait analysis, captured by

wearable sensors or depth cameras like the Kinect, has been used to detect wandering patterns[8, 4], prescreen for MCI[17], and differentiate MCI from healthy controls based on oval-path walking.[16]

2.13. Wearable Technologies in Alzheimer's Care

Rapid evolution of tech in the wearable segment from simple pedometers to very complex smartwatches and bio sensor bands which in turn has brought about a transformation in health care monitoring. Now physiological monitoring is a part of the fabric of daily life with devices like the Fitbit, Apple Watch and Empatica E4. We see in this area of smart wearables that they provide objective info on activity levels, sleep quality, stress response, and heart rate variability which in some cases also play a role in the progress of disease in Alzheimer's research.[5] Also we have longitudinal tracking which is a result of continuous noninvasive data collection which in turn see clinicians monitor for subtle cognitive and behavioral changes over time as opposed to random clinic visits. Also with that these smart wearables give care givers a sense of control by reporting on atypical patterns which also include wandering and falls.[15, 1]

2.14. Limitations and Data Challenges

They are in fact what we see as deficient in terms of accuracy, data integration, and user compliance which at the same time they claim to achieve.[2] Also we see that reliability is put to the test by sensor drift, noise in the data, and subject variability. In the case of long term use in the elderly population battery life, comfort, and aesthetics also play very large and critical roles. We see that these devices which constantly send health data which is very sensitive and requires a secure encryption policy also have issues with data privacy. To address these issues out there researchers are reporting in to implement AI right on the device for real time analysis which does away with the need to send data to the cloud.

2.15. Integration of AI and Wearable Systems

The focus of modern health research is on multimodal fusion, which is when data from different types of sensors (acceleration sensors, EEGs and ECGs) is brought together in order to maximize predictive ability.[6] Using different types and sources of modalities allows for different uses with some modalities supplementing each other.

For example, the mobility deficit measured through an accelerometer could be used with an EEG for measuring brain activity, and using a microphone to evaluate cognitive-linguistic behavior.[9] Potentially successful fusion models using DNN or GNN could be developed to integrate multiple signals and provide a comprehensive diagnostic decision. A specific example of this is the use of gait and heart rate variability to distinguish between mild cognitive impairment (MCI) and normal aging, which has been found to outperform the results of either sensor stream when used alone.[13]

The next generation of multimodal fusion can be seen in the NeuroGuard system that integrates wearable, behavioral and voice data to allow for more intelligent and personalized monitoring.

2.16. Real-Time Processing and Edge AI

Practical implementation of solutions requires real time decision making. Cloud systems are at a standstill with issues of latency, connectivity, and security. Into this comes Edge AI which we see as a revolutionary solution.[18] At the edge models are deployed directly to wearable devices or local systems which is what makes it so revolutionary. Edge computing we see to have low latency, data privacy and offline features which is why it is perfect for at home Alzheimer's care. NeuroGuard we put forward as a model which processes data locally but also has secure cloud back up for physician access.

2.17. Assistive Voice Technology in Cognitive Care

Amazon Echo, Google Home and Siri as we see it today have brought natural language interaction with tech into the mainstream. In health care we are seeing more and more of these systems used to send out reminders to patients to take their meds, to alert them to safety issues, and for social interaction all of which is very much the care setting for Alzheimer's patients that have issues with disorientation and memory loss.[7] We also see from studies that these

types of agents may in fact reduce the burden on caregivers and improve patient' adherence to routine.[3] At NeuroGuard we have added to this by introducing an AI powered conversational agent which we have trained specifically for cognitive support.

Summary of Research Gaps

The literature reviewed highlights a disjointed world in which detection of Alzheimer's, wearable monitoring, and assistive support are considered distinct spaces. Potential has been shown in each space through studies, but without scalability into one system.[2, 12, 7]

The main research gaps are:

- The fewer common frameworks that marry wearable, behavioral, and clinical biomarkers.
- Sparse attention to on-device processing by AI algorithms for real-time analysis preserving privacy.
- No personalized, adaptive voice-based support for caregivers and patients.
- Limited clinical verification of multimodal, AI-based systems under real-world settings.
- Through an integrated, multimodal, and user-oriented solution NeuroGuard solves these issues in the next stage in proactive Alzheimer's care.

3. PROBLEM STATEMENT

3.11. Background and Context

Alzheimer's Disease (AD) is a progressive brain disorder that slowly destroys memory and thinking skills and eventually the ability to carry out the simplest tasks. It occurs largely in older people, and by the later stages they are completely dependent. The incidence of dementia worldwide is over 55 million people who suffer with it, and approximately 70% of these is Alzheimer's Disease. This number will dramatically rise over the coming years because of the number of ageing people and is estimated to reach over 139 million by the year 2050. In India it is estimated that approximately 8 million persons are suffering from a dementia related illness, the great majority of them undiagnosed and or only diagnosed very late.

Although the disease is quite serious, it is not usually diagnosed early. The early symptoms are small memory disturbances, confusion, and disorientation. These are regarded as normal for old age, and so it is when the disease becomes moderate or severe that the doctor is consulted. By this time, brain degeneration has taken place, and so is nearly useless as a treatment. The tests now in use, namely, the RMI [magnetic resonance imaging test], the PET [positron-emission tomography] scan, and the examination of the fluid in the cerebral cavity, do show very nicely the existence of the disease, but they are very expensive, uncomfortable, and needed are elaborately at present carried out with a great deal of skill.

In addition to this, they merely tell what is the condition of affairs at that time, and not the continuing changes in the brain. For this reason, the opportunity for early diagnosis is very often lost. It is now made possible by the means that have been found for the study of brain, and the body indicators, through the use of wearable devices, biosensors, and artificial intelligence (AI). With the continued use of the data from the wearables use of the work done by the assistance and chemical analysis of these devices, the brain can be examined frequently, and changes determined. If the data from these wearables is connected with the results of AI, the pattern which indicates early change, and hence a possibility of Alzheimer's, can be found. This information has been gathered under the system called NeuroGuard, a wearable apparatus based on AI, which is able to find early indications of Alzheimer's, and also to assist, through forms of helping the patient, in safety and the emotional state.

3.12. Challenges in Current Alzheimer's Management

There are three major areas of multifactorial difficulty for which Alzheimer's attacks are more possible. Diagnosis, observation, assistance. The attention that has been expended upon, and more particularly, various inventions rabbits has been made to aid gamely in these matters, to which at present the holds the better.

3.12.1. Challenges in Current Alzheimer's Management

The current challenges in Alzheimer's Management are:

Delayed Diagnosis: The early diagnosis of various cognitive disabilities is frequently unrecognised regarded as a part of the normal aging process. By far the greatest incidence of error occurs from delayed diagnoses. Low cost of the tests required.

Inexpensive Testing: Neuro-imaging been now carried out, and biomarker tests for the diagnosis of alterations of the brain of which there is a great amount improvement but are expensive and beyond most complications of the general public particularly found.

Involvement of Specialists: Specialized neurologists or psychiatrists must make the appropriate diagnosis, which is a problem where resources of health are scarce.

No Objective Measurement: Most standard measures rely on the subjective description of behaviour or the survey method that would fail to reveal the existing minor symptoms of the insipient degenerative disease.

3.12.2. Monitoring Challenges

- **Fragmentation:** The present means of examination monitor a single factor only, e.g., activity or pulse rate, and do not take into account the pathophysiological data e.g., neurological and emotional and physiological.
- **Non-continuous:** The patient is not continuously examined but at stated intermittent periods, and therefore the fluctuations and prodromes are lost.
- **No Dynamic Disease Curve:** There is the very least power of collaboration or forecast dynamic in the progress of disease from successive follow-up examinations.

3.12.3. Support and Safety Issues

- **Reduction in Autonomy:** As the confusion increases and memory goes the mistaken one becomes absolutely dependent on the helper in every department of everyday life.
- **Safety Dangers:** Wandering, falling, and confusion in the way are dangerous experiences, especially in those who live alone.
- **Isolation:** The patient is prone to be irritated, anxious and lonely, which increases the symptoms.
- **Care Giver Problems:** Family suffers financially and emotionally such heavy burdens because of the necessity to give continuous observations and supervision.

3.13. Identified Research Gaps

The most recent scientific research and technical advancements have identified several areas of discrepancy between research and real-world Alzheimer's prediction systems:

- **Unidisciplinary Focus:** Most existing systems predict using just one type of sensor or recorded data (like EEGs or position tracks) while, Alzheimer's disease has a unique signature that includes multiple sources of physiological, emotional and cognitive information.
- **Short-Term Data Emphasis:** Many existing scientific studies have been done with a short timeframe (days or hours) – which does not accurately reflect the progression of a disease like Alzheimer's that develops over many years.
- **Personalization:** Existing models have been created using the average behaviours of individuals without accounting for the behaviors of the specific individual based on their normal activities in their community and/or in comparison to their genetics.
- **Lack of Intelligent Assistants:** Most present day detections only carry out symptom recognition, without supplying help, reminders, and emotional assistance.
- **Security of Data:** There are serious security issues relating to the gathering and transmitting of sensitive health data formulation of methods for their resolution is an essential for facilitating widespread deployment. The clear filling of these voids in knowledge is the pivotal motivating force of the NeuroGuard project.

Filling these knowledge gaps is the core motivation of the NeuroGuard project.

3.14. Statement of the Core Problem

In the light of present difficulties and inadequacies, the core problem can be defined as the following. There is an urgent need for a holistic, low-cost, and intelligent wearable system that will monitor, detect early, assist continually applicable to Alzheimer's subjects in the real world.

This problem invites three interdependent goals:

- **Early Detection:** To use the information from the multi-modal biosensor measurements and predict the probability of occurrence of Alzheimer's in early phases.
- **Stage Detection:** To forecast the present and future stages of the disease as mild, moderate, or severe applying standard statistical and machine learning algorithms.

Assistive Facilities: Online services to provide adaptive reminders, fall detection, emergency messages and emotional interaction will be offered in real time through a range of devices.

The issue is defined not just as a technical problem but rather a health care need that focuses on individuals as well as.

3.15. Need for the Proposed System

The NeuroGuard system was specifically developed to address these issues using a combination of Internet of Things (IoT) technology, embedded systems, machine learning and AI-enabled voice assistance. In addition to being a tool for diagnosing diseases and disorders, the NeuroGuard acts as an assistant for people with neurological conditions around the globe by collecting real-time data regarding a patient's health, predicting the likelihood that a patient will develop a disease, and assisting a patient with his/her daily activities.

The following are some examples of the specific problems that the NeuroGuard will address:

- **Early Risk Identification:** Continuous monitoring of physiological and emotional data through biosensors is able to detect early signs of physiological and emotional impairment associated with Alzheimer's disease.
- **Cost-Effective Screening:** The means of screening that is cost effective and available in any geographic area as compared to hospitals.
- **Proactive Diagnostic Support:** Use of current analytic data combined with historical analytic data are used by healthcare providers and family members to support proactive prevention of diseases through early intervention.
- **Safer Experience:** Integrated emergency communications, GPS location services, geofencing, and fall detection technologies will provide patients and caregivers with confidence that if an emergency occurs, they will receive assistance.
- **Emotional Security:** The integrated AI assistant provides companionship and assistance with memory through verbal reminders and cueing, thereby lessening feelings of anxiety and disorientation.
- **Individual Adaptive Learning:** Through the ability to learn over time, the AI assistant will predict better than any traditional form of health care and provide an individualized level of advice through an individualized pattern of use.

Through the use of much of the same technology and methodologies that traditionally have been used in institutional-based care for Alzheimer's disease, the NeuroGuard is able to transform how Alzheimer's care is provided from the reactive care traditionally provided in a hospital setting to providing a proactive, home-based, and patient-centered model of care.

3.16. Scope and Expected Impact

In part, NeuroGuard redefines the treatment centre-based, reactive model of Alzheimer's care to a proactive, home-based experience, patient-centred model of patient treatment.

Area and Impact

The area of the NeuroGuard project is both medical and social. It is at first a pre-screening, stage-foresight strategy for Alzheimer's but as a modulated platform it is also capable of scaling to other neurodegenerative and mental disorders, such as Parkinson's, dementia, and cognitive impairment by reason of aging disabilities.

Prioritized Early Diagnosis and Treatment - Shortening the time span from the onset of symptoms to diagnosis.

Less Pressure on Caregivers - Automation of monitoring and alerts to lessen the stress on the family. Improvement of Independence on the Part of the Patient - Self-management facilitated via reminders and voice support systems.

4. PROPOSED SOLUTION

4.11.Introduction and Overview

Alzheimer's Disease (AD) known as a world health systems disease causing loss of cognitive, memory and functional impairment. It results in delay of diagnosis, and early participation has little effect. Furthermore down the line, patients need looking after when it comes to issues of safety, such as falls, and again that responsibility is transferred to families. Current methods include tests such as CSF analysis, which expensive imaging such as MRI, or any type of evaluation from any clinics that fail to reflect the differences and evolution of the condition. It leads to monitoring in some intervals, which is not safe periodical checks. NeuroGuard is a smart wearable AI-based product that provides early detection and assistance for Alzheimer's afflicted individuals in order to highlight the crucial gaps in cognition and health.

4.11.1. Core Functionalities

NeuroGuard is designed around two stands:

1. Early Detection: Continuously gathering and analyzing physiological and behavioral data (EEG, EDA, motion, location, and temperature) with AI to detect patterns of early cognitive symptoms (Mild Cognitive Impairment - MCI) or differences between stages (Mild, Moderate, Severe).

2. Assistance & Safety Monitoring: will have all of the real-time alerting, and functions for improving both patient safety and support with activities of daily living. This will include detection of fall or wandering through geofencing, activity or sleep pattern monitoring and maybe able to provide reminders or cognitive aides through mobile application.

4.11.2. Objectives of the Proposed Solution

- To develop a comfortable and non-intrusive wearable device capable of collecting relevant multimodal physiological and behavioral data over extended periods.
- To design and implement a robust data pipeline for secure transmission, storage, and processing of sensitive patient data.
- To build, validate AI models which are capable to classify AD stages (including a healthy Control baseline) with high accuracy (greater than 90%) based on the collected time series sensor dataset.
- To add functionality for monitoring of real-time activity on the device (fall detection/wandering alerts).
- To design intuitive mobile application interfaces for Patients (easy access to help) and Caregivers (full monitoring/alert management).
- To comply with strict standards for protecting the privacy and security of users' information (i.e. Hipaa compliance guidelines).

4.11.3. Target Users

NeuroGuard is beneficial to:

- Those who are at risk of or have been diagnosed with cognitive disorders (early detection of AD).
- Those diagnosed with AD (monitoring of stages of AD).
- Family members or other types of care providers (to provide peace of mind, reduce the burden of monitoring, receive timely notifications, and gain insight into how the individual is doing).
- Clinicians (as an adjunctive tool that provides objective, ongoing data to assist in making a diagnosis and tracking disease progression, and possibly also used to evaluate the effectiveness of treatment).

Our goal is to transition from episodic (reactive) care to a system of *proactive* ongoing monitoring, with support that improves care to people with AD while decreasing the burden on their families.

4.12. System Architecture

NeuroGuard engage an architecture consisting four main components, designed for collection, processing, analysis of data and user interaction efficiently:

- 1. Sensor Unit (Wearable Device):** Primary data acquisition component which will be worn by the patient.
- 2. Companion App (Mobile Application):** It acts as data gateway, user interface, running on a smartphone (iOS or Android).
- 3. Cloud Platform:** Secure, scalable cloud-based infrastructure for the storage, processing of data, and hosting of AI models.
- 4. AI Core:** Central intelligence engine which will be hosted on the cloud platform is responsible for analyzing the time series sensor datasets and will generate alerts.

4.12.1. High-Level Interaction Flow

- Wearable Device is continuously collecting sensor datasets (EEG, ECG, EDA, Temp, GPS).
- Data is periodically transmitted via Bluetooth Low Energy (BLE) to the paired Companion App on the patient's or caregiver's smartphone.
- The Companion App buffers the data and securely uploads it to the Cloud Platform via cellular or Wi-Fi connection whenever available.
- In cloud platforms, raw data goes through pre-processing and extraction of features.
- The AI Core analyzes the extracted features for classification and detects any anomalies (for example, detecting potential falls or indicating if a patient has wandered).

The AI generates results, insights and alerts that are sent to the cloud platform from which they can be accessed via the Companion App.

The Companion App helps provide caregivers with important information and alerts, and has features such as reminders and navigation to get to their patients based on preset configurations.

The architecture we built is balanced to accommodate device requirement constraints such as power and compute capabilities while providing fault tolerant, scalable data handling using AI.

The mobile application is also important in that it provides the link between the caregiver (user) and the application.

4.13. NeuroGuard Sensor Unit (Wearable Device)

A wearable sensor is at the core of the NeuroGuard system; therefore, this block can continuously collect health data without discomfort. The design ensures it's simple to wear, has a long battery life, is accurate and can be used by anyone.

4.13.1. Form Factor

This is a combination approach:

- **Primary Wristband:** It functions similar to a smart watch or fitness band. It has a main processor, sensors for movement, sensors for heart rate and pulse, sensors for stress and temperature, Bluetooth for wireless data transfer and battery. It is made light and comfortable for every day wear.

- **Optional Headband or Ear Clip:** A small band or clip, easy to wear, which can read brain signals from some special sensors. This device connects wirelessly to the wristband and helps to record the brain activity without having to keep everything in a bulky device. This is a more convenient system to wear, and can be used in sleep, as well as when active. During the experiment both with the sensors built in the ear area will be experimented on.

- **Built-in GPS:** The wristband also has a GPS unit to track the wearer's location when needed.

This setup has many parts, so that every sensor can be placed where they will allow for the greatest effect, such as the brain sensor which is on the head, and the stress sensor, that is on the wrist. This is done to keep the user in comfort and wearability and not a cumbersome device. This system continually gathers body vitals of all types, and has the brain sensing device, that has more information dependent on the more in-depth data about the mental state and nerve activities about the individual where it is worn.

4.13.2. Integrated Sensors

A Raspberry Pi 5 Model centralizes the continuous sensor array of biopotential sensors (BioAmp EXG Pill: EEG/ECG), pulse oximetry and heart rate (MAX30102 PPG), electrodermal activity - GSR measure, inertial measurement unit - MPU6050 (3-axis gyroscopes), skin temperature (MLX90614 ESF) and GPS Module (NEO-6M) data.

- **BioAmp EXG Pill (EEG/ECG):** A biopotential sensor that can be used with dry electrodes and snap cables to obtain electrical signal output from both the brain (EEG) and heart (ECG). By using the ADS1115 Analog-to-Digital Converter (ADC), these device signals are converted into digital format for data gathering and monitoring purposes, such as for concentration, sleep cycles, or alertness.

- **MAX30102 (Photoplethysmography - PPG):** An integrated pulse oximeter and heart rate monitoring module powered by multiple optical sensors to record blood oxygen levels and heart rates continually. The raw data collected can be used to calculate heart rate variability (HRV), which allows for assessing physical and emotional levels of physiological stress and assessing overall cardiovascular health.

- **Grove GSR Sensor (Electrodermal Activity - EDA):** The sensor measures galvanic skin response (GSR) to assess the degree of electrical conductance of one's skin. Such readings provide an indication of the activity of the sympathetic nervous system and physiological arousal or provide a representation of stress, excitement, and/or emotional surprise.

- **MPU-6050 (Inertial Measurement Unit - IMU):** A 6-axis motion tracking sensor including a 3-axis accelerometer and a 3-axis gyroscope for measuring movement, orientation, and balance. As such, this sensor can be used to collect movement, detect patterns of walking (gait), and detect sudden falls or loss of balance.

- **MLX90614 ESF (Skin Temperature Sensor):** An infrared (non-contact) thermometer measuring the skin surface temperature. This sensor allows for continual measures of small temperature variations in association with changes in stress, daily physiological cycles, and an overall impact on health without having to touch the skin to obtain a measurement.

- **NEO-6M GPS Module (Global Positioning System):** It is a type of Global Positioning System (GPS) that tracks the position and movement of an individual in an outdoor setting, providing real time position information for distance travelled; supports geofencing (i.e., where an alert will be sent if the user moves out of their specified safe area).

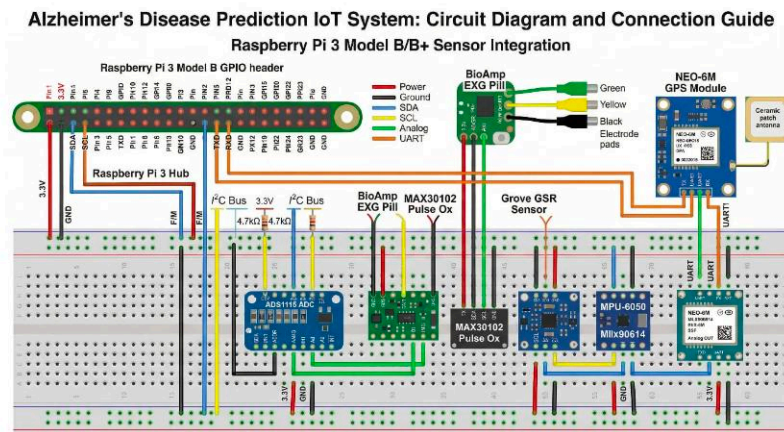


Figure 1: Circuit Diagram of NeuroGuard

4.13.3. Processing and Communication

- **Microcontroller Unit (MCU):** A low power ARM Cortex-M based MCU in the wristband, dedicated to sensor interfacing, basic data buffering, timestamps, possible on-device preprocessing (e.g. filtering noise) and BLE comms management. Edge computation capabilities will be investigated for simple jobs such as threshold based fall detection to allow for quicker alerts even if the phone is disconnected.
- **Communication Module:** Bluetooth Low Energy (BLE 5.0 or better) for low power wireless comms with the Companion App

4.13.4. Power Management

- **Battery:** Rechargeable Lithium Polymer battery sized for an estimated 24 - 48 hours continued operation (with GPS usage massively affecting this). There is a target of at least one daily cycle of night/day between charges.
- **Charging:** Simple magnetic pogo-pin or inductive charging interface to allow easy use by patients/carers.

4.13.5. User Interface Elements

- **Physical Button:** A singular, easily pressable button that acts as a panic/help button for patients to manually trigger an alert to the carer via the companion app.
- **LED Indicators:** Simple LED indications of power status, charge status and Bluetooth connectivity status.

The hardware design will be fundamental, robust, water resistant (for daily activity) and be made from biocompatible materials suitable for continuous skin contact.

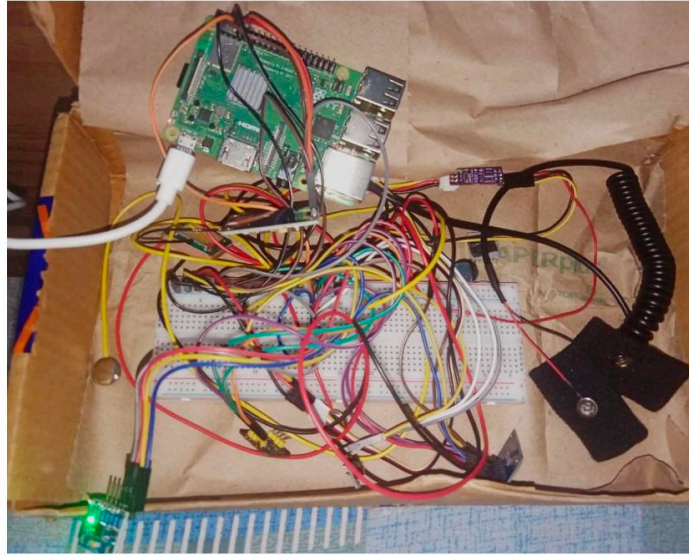


Figure 2: Hardware Prototype of NeuroGuard

4.14. NeuroGuard Companion App (Mobile Application)

The congratulatory Mobile App is in between the wearable device, cloud platform, and user. It serves as an interface for the user, manages data flow, and provides assistance.

4.14.1. Role and Platform

- **Data Relay:** This establishes and maintains a secure Bluetooth Low Energy (BLE) connection to the NeuroGuard Sensor Unit; streams (real-time) or receives batches of Sensor Data; stores them locally and uploads them securely to the NeuroGuard Cloud Platform (<https://www.neuroguard.com>).
- **User Interface Hub:** There are two separate user interface hubs (one for the Patient and one for the Caregiver) offered by the Mobile App.
- **Assistance Delivery:** The Mobile App will allow delivery and/or display of Alerts & Reminders and other Cognitive Aids.
- The platform (Mobile App) will be designed and built natively to work on both iOS and Android platforms allowing for maximum user access.

4.14.2. Data Synchronization

The Mobile App constantly searches for the wearable device to connect (e.g., via BLE). When the Mobile App has connected to the wearable device, it negotiates whether or not the data should be transmitted either: (1) as real-time streaming (for critical data such as IMU during Fall Detection Analysis), or (2) as batches of data (for historical HR, EEG records), thus conserving battery usage on the wearable device. The Mobile App timestamps each data record upon receipt, and stores the data in a Local Database temporarily on the Smartphone. After data is buffered, the Mobile App will try to transmit the buffered data to the NeuroGuard Cloud via Wi-Fi or cellular network. Data will be transmitted in batch form and in encrypted format. If a connection is not available, the buffered data will remain locally on the Smartphone until the connection is restored preventing the loss of data.

4.14.3. Patient Interface (Simplified)

- **Features of a Simple Design:** Easy-to-see buttons, easy-to-read text, very distinct colors, and easy-to-use navigation assist people who might have difficulty thinking clearly.
- **Basic Information:** A basic form of data will indicate the current date & time; the amount of battery remaining in the device; whether or not the device is connected to a smart phone, computer, router, etc.
- **Examples of Helpful Features:** (It is important that the caregiver be the one who programs these features).

A reminder through sound and vision of when someone needs to take their medication, see their doctor, etc.

A simple map showing the location and a reference point of the location from the home (to be done by the family caregiver). The use of the map with spoken instructions for how to get there.

A camera attachment, which could be used to help remember people by name or who they are related to. This must be programmed and given permission to be used by the caregiver.

There will be visual feedback when the wearable device's panic button is pressed.

A focus on the passive nature of the device as being limited in the amount of interaction required by the recipient of the services.

4.14.4. Caregiver Interface (Comprehensive Dashboard)

- Secure Sign-in (to get patient information)
- Home Screen
 - Overall Status: List of current patient area (their location), latest heart rate, level of physical movement (at rest, walking, active) and how much battery is on their provided wristwatch or other wearables.
 - Alerts: All active alerts will be on this page (if a patient was detected as having fallen, if they left a pre-defined area, if they used the panic button).
 - Summary of Key Trends: Summarization of activity level, hours of restorative cyclic or non-cyclic sleep, and heart rate or heart rate variability averages from the prior 24 hours or 7 days.
- Historical Data Exploration/Access: Graphs and summaries of the sensors' data over periods of time of your choosing (1 day, 1 week, 1 month) to determine trends and/or patterns related to a specific patient's past.
- Configuration of Parameters including:
 - Geofencing (defining a safe zone for the patient/defining an alerting method when they leave the defined zones)
 - Reminders (setting up reminders for taking medications, appointments, tasks, etc.)
 - Emergency contact information (creating &/or editing)
 - Alert preferences (the method in which the patient wants to receive/receive alerts; push notification, short text message).
- AI Insights: Display of information (i.e., how AI is determining the categorization of cognitive impairment/stage: classification based on neural and physiologic signals exhibited, will exhibit changes over a period of time, and identifying any excessive variations or deviance to normative population statistics).

4.14.5. Offline Capabilities

The application is designed to continue gathering information from the wearable device without needing an internet connection, and then locally store that information for later use. While offline, there are some core safety features like geofence boundary verification (if the mapping data is pre-loaded) and a limited capacity to have fall detection logic (if that feature is available on the phone and/or the wearable device). If a panic button were activated on the wearable, the app will still be capable of notifying an appropriate caregiver through a BLE connection.

4.15. NeuroGuard Cloud Platform

Our cloud platform provides backend support for all your data generation from wearables and backend support of the mobile application itself, including scalability, security, performance, and stability.

4.15.1. Purpose

- **Data Ingestion:** Receiving data securely and validating incoming data streams coming from multiple Apps.
- **Scalable Storage:** Store vast amounts of time-series sensor data and derived features persistently and cost-effectively.
- **Data Processing:** Execute pipelines for cleaning, normalizing, and transforming raw sensor data into features suitable for AI analysis.
- **AI Model Hosting & Execution:** Provide the computational resources to run the NeuroGuard AI Core models for detection and assistance.
- **API Services:** Offer secure Application Programming Interfaces (APIs) for the Companion App and potentially future integrations (e.g., clinician portals, research platforms).
- **Alerting Infrastructure:** Manage the generation and delivery of alerts based on AI Core outputs or rule-based triggers (e.g., geofence).

4.15.2. Components

- **API Gateway & Balancing Load:** Managing incoming requests , helps in users authentication, and traffic distribution across backend services.
- **Data Ingestion Service:** Receives data batches, performs initial validation, and queues data for processing.
- **Time-Series Data:** Database which is optimized for storing and querying sensor data which will have a timestamp with it(e.g. imescaleDB, AWS Timestream). The platform has been set to accept time series queries against it. The metadata for each query is maintained in a NoSQL database. To process data we will implement a data processing pipeline on AWS Lambda, Google Cloud Functions, or Kubernetes. The following are some things that we will achieve through use of the above cloud hosting:
 - **Perform Data Cleaning:** When performing data cleaning, we will remove missing data points, and produce filtered data from any noise present in our data (for example, we can filter out motion artifacts in EEG data)
 - **Perform Data Normalization/Standardization:** This means processing and scaling the data to provide a consistent scale across multiple devices.
 - **Perform Feature Extraction:** We will calculate with your data various features such as heart rate variability (HRV) measures, EEG band power averages and spacial-temporal gait characteristics (e.g., stride length, cadence, and gait variability), and activity counts, and also calculate estimates of sleep stages, and have created life space mobility indices.
 - **AI Model Serving:** The AI model serving platform is the engine that runs, hosts and scales your trained ML models. In addition, the AI model serving platform provides version control of the AI model used, and also provides real-time inference.
 - **Alerts System:** Our alerts system works with AI outputs from the AI model serving platform to create rules, format notifications, and deliver alerts via push notifications and/or SMS messages to the App.

4.15.3. Security and Privacy

- **Compliance:** Designed from the beginning to incorporate HIPAA (or other pertinent laws like GDPR).
- **Encryption:** Data is safe from prying eyes during transfers (via TLS/SSL) and when stored (encryption at the database level).
- **Access Control:** Each user can only view data as per their assignments which is classified as Role-based Access Control (RBAC). On top of this, access is tightly regulated.

- Anonymization/Pseudonymization: Where possible, we apply techniques to remove identifying details from the data used to train models or for other research.
- Secure APIs: All API endpoints are subject to authentication and authorization.
- Regular Audits: Routine Security audits including Penetration testing are integrated. Cloud systems are built with inherent, scalable, reliable, and secure characteristics.

4.16.NeuroGuard AI Core

AI is the intelligent engine for NeuroGuard. producing the early diagnosis and progress giving sophisticated aide using complexed sensor and other data. It is made up of various specialized machine learning and signal processing modules.

4.16.1. Early Detection Module

- Objective: To determine where (Control, Mild Cognitive Impairment, Moderate Cognitive Impairment or Severe Cognitive Impairment), from the sensor data, the patient is at present, and also to identify significant changes over time as potential markers of disease progression.
- Methodology: We utilized a supervised machine learning approach that included the use of ensemble models for improved and accurate performance. Given the success of our preliminary results (>98% accuracy), we have utilized a well-tuned LightGBM (Light Gradient Boosting Machine) model as our primary classifier, and are now exploring hybrid methods which may include Recurrent Neural Networks (RNNs such as LSTM) or 1D Convolutional Neural Networks (CNNs) for temporal feature extraction from the raw/near raw signal segments (i.e., EEG, IMU) prior to inputting them into the LightGBM model.
- Input features: A comprehensive set of features was created from the raw sensor data streams over predetermined time windows (i.e. hourly, daily, nightly): EEG – Power Spectral Density (PSD) data within standard bands (Delta, Theta, Alpha, Beta), as well as ratios between these bands(e.g., Alpha/Beta ratio, Theta/Delta ratio), peak frequencies as well as Entropy measures. HRV – Time Domain metrics (RMSSD, SDNN, pNN50) and frequency domain metrics (LF/HF ratio). EDA – Mean SCL and number of SCRs and amplitude of SCRs.– Gait (IMU): Gait speed, stride length variability, cadence, step count, symmetry measures (calculated during walking segments identified by activity classification).
- Activity (IMU): Total activity counts, time spent in different activity levels (sedentary, light, moderate), fragmentation of activity.
- Sleep (IMU/EEG/HRV): Estimated total sleep time, sleep efficiency, wake after sleep onset (WASO), time spent in different sleep stages (if EEG allows reliable classification), sleep fragmentation index.
- Skin Temperature: Mean temperature, circadian temperature rhythm amplitude and phase.
- GPS: Life-space mobility area (e.g., maximum distance from home), time spent outdoors, regularity of movement patterns.
- Engineered Features: Ratios (EEG Alpha/Beta, EEG Theta/Delta), Accel Magnitude statistics.

4.16.2. Model Architecture (Primary - LightGBM)

- Gradient-Boosted Decision Trees deliver high performance for tabular data as well as computational efficiency and are able to deal with a number of different types of features.
- Hyperparameters (n estimators, learning rate, num leaves, reg alpha, reg lambda etc.) will be optimally tuned using GridSearchCV or RandomizedSearchCV and stratified cross-validation, as appropriate, based on the need to achieve >98% accuracy performance.
- The parameter class weight='balanced' or techniques such as SMOTE (as implemented in the advanced training script) will be important when dealing with the intrinsic class imbalance.

Output: The model will output a probability score across the four classes (Control, Mild, Moderate, Severe) with the maximum score given as the primary class prediction. The predicted class will output a level of confidence. Weekly

or monthly trend analysis will be employed to examine brain change to support ongoing stage progression, providing the caregiver/clinician with a notification rather than any sort of definitive statement.

Training & Validation: Initially trained on the high quality labelled clinical studies dataset for iterative re-training, fine tuning and rigorous validation against clinical diagnostic state (using metrics of accuracy, precision, recall, F1-score, AUC-ROC for each class).

4.16.3. Progressive Assistance Module (Real-time/Near Real-time Analysis)

- **Fall Detection:**

- **Algorithm:** Primarily based on IMU (accelerometer and gyroscope) data. A combination of thresholding (detecting high impact magnitude and sudden change in orientation) and a lightweight machine learning classifier (e.g., SVM or a simple neural network trained specifically on fall vs. non-fall movements) running potentially on the wearable or phone for low latency.

- **Logic:** Detects characteristic impact patterns followed by a period of inactivity. Triggers an immediate high-priority alert to the caregiver via the Companion App.

- **Wandering Detection (Geofencing):**

- **Algorithm:** Rule-based system using GPS location data.

- **Logic:** Compares the patient's current GPS coordinates against predefined "safe zones" configured by the caregiver in the Companion App. If the patient's location moves outside a defined zone, an alert is triggered. Alert logic may include delays or require confirmation of sustained exit to reduce false alarms. GPS accuracy limitations in certain environments (indoors, dense urban areas) are acknowledged.

- **Activity & Sleep Monitoring:** – **Algorithm:** Uses signal processing and Machine Learning to monitor the IMU signals to classify activity types

(rest, walking, exercise) and sleep parameters such as sleep duration, sleep efficiency and sleep interruption

detection. The EEG if available enhances the ability to classify sleep stage. – **Logic:** Total daily or weekly activity, steps per day, as well as sleep patterns are computed on a daily basis This

information is communicated to the caregiver as trends indicating normality/abnormality. Significant deviations from baselines (ex drop in activity level, highly fragmented sleep) might lead to infrequent follow-up reminders for the caregiver to check on the patient.

- **Future AI Features (Exploratory):** – **Agitation Prediction:** Research shows relationships between physiological factors (Heart Rate Variability,

Electrodermal Activity) and episodes of agitation or anxiety which are common in AD. Models could help predict probability of attacking events allowing for productive intervention.

- **Personalized Reminders:** Reminder timing or modality could be altered based on AI assessment of cognitive state or activity level of patient. (ie remind only if patient was sedentary).

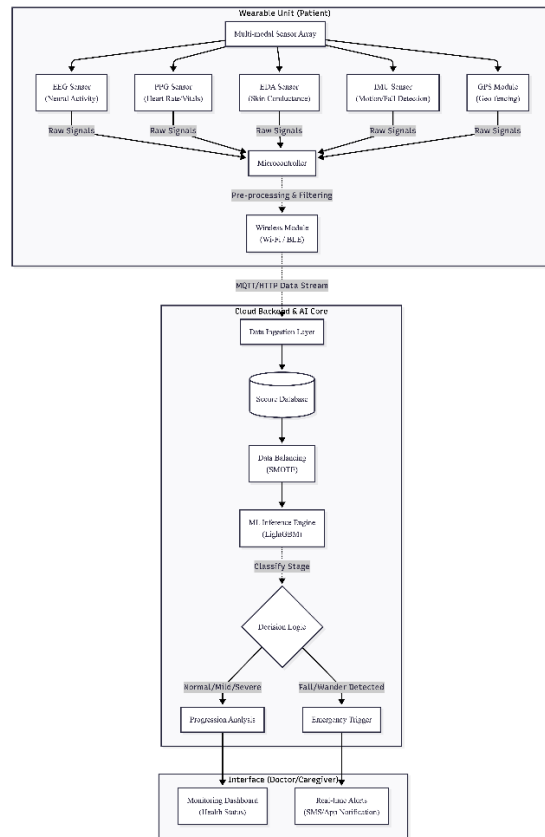
AI Core represents the neural (brain) of NeuroGuard which would continuously learn and modify itself in order to give personalized, timely, useful information for both early diagnosis and ongoing patient well-being. Development will be continuous requiring extensive data and cooperation with clinical professionals.

4.17.Data Flow and Interaction Scenarios

It is important to understand the flow of information as well as how the users will interact with the system in order to best express the applications.

4.17.1. Detailed Data Journey

1. Sensing: Sensors of the NeuroGuard Sensor Unit (EEG, PPG, EDA, IMU, Temp, GPS) continuously record raw physiological and behavioral information.
2. On-Device Buffering/Processing: MCU of the wearable times, buffers, and applies minimal pre-processing (e.g., simple filtering). On-device fall detection trigger potential.
3. BLE Transmission: Data packets are securely transmitted via BLE to the associated NeuroGuard Companion App on the mobile phone. Frequency and batching strategy optimized for power.
4. App Reception & Buffering: The Companion App receives the information, checks its integrity, buffers it temporarily to a local database, and waits for network availability.
5. Secure Upload: As Wi-Fi or cellular connectivity is established, the app uploads the buffered batches of data to the ingestion service of the NeuroGuard Cloud Platform encrypted via HTTPS.
6. Ingesting and queuing information in the cloud: the information comes into the cloud platform where it is authenticated, validated by format, and queued up to be processed.
7. Processing pipeline: Check out the data from the queue, clean it and normalize it, along with extracting critical features (HRV, PSD, Gait Parameters, etc.). The raw and processed information is stored and organized by time in a time series database.
8. AI analysis: Scheduled tasks/triggers send the extracted features for the AI Core models (ex- LightGBM trained Classifier for stage assessment and falling/wandering algorithms) based on the timeframes in which the feature was collected.
9. Generating insights and alerts: AI Core generates predictions (the likelihood of stage) detects anomalies (like potentially falling or breaching geofences), or summarizes patterns (sleep quality and activity levels).
10. Cloud database notifications: Results are stored in the cloud database; if the condition of high-priority falls/wandering or significant amount of changes occur, the alerting service is triggered.
11. Push to app: The alerting service securely pushes notifications/messages to the Companion App of the caregiver using the provider notification service network or SMS Gateway. Regular insights (like daily summaries) will be pulled down to the app via API when active.
12. User engagement: When caregivers receive alerts/notifications on their mobile device, they will see real-time data, historic trends and AI insights through dashboard of the app. Patients will receive reminders and/or use the assistant features through a simple interface.



Flow Diagram of NeuroGuard

Key Interaction Scenarios

Scenario 1: Fall Detected

1. The wearable IMU recognizes a strong impact and loss of movement afterwards.
2. The wearable sends an urgent BLE signal to the phone immediately (or activates an internal alert if the phone is not connected).
3. The Companion App gets this signal, verifies that a fall has occurred (it may also be using the phone's sensors), and sends an urgent alert to the Cloud Platform.
4. The Cloud Alerting Service sends an alert with geolocation (if available) to the Caregiver's Companion App and any other designated emergency contacts.
5. The Caregiver gets notified of a fall for [Patient Name] with a location/time stamp.

Scenario 2: Patient Wandering

1. The wearable GPS sends location data to the Companion App.
2. The Companion App sends location data to the Cloud Platform.
3. The Cloud Platform checks the geolocation against the Caregiver defined geographic limit.
4. The patient has remained outside of their defined geographic limit for a set period of time (e.g., over 2 minutes).
5. The Cloud Alerting Service sends a wandering alert to the Caregiver.
6. The Caregiver receives a notification for [Patient Name] that they have left their defined geographic limit. Current location is [Map Link/Address].

Scenario 3: Potential Stage Advancement Signals Indicated

1. The AI Core reviews longitudinal data (for example, over the course of a number of weeks).
2. Consistency in changes of pattern has been detected within the model (for example, reductions in Alpha/Beta ratio, increased gait variability, and worsening sleep fragmentation), which results in increased probability of more severe stages.
3. AI Core creates a low-priority insight/alert.
4. The Cloud Alerting Service will send a notification to the caregiver's mobile app.
5. The Caregiver receives a notification stating: "Insight: NeuroGuard Analysis indicates probable changes in [Patient Name]'s cognitive indicators in the past [Time Period]. Consider reviewing patterns or speaking with a clinician". (Note: avoids definitive diagnostic statements).

Scenario 4: Patient Utilizes Cognitive Support

1. Patient feels disoriented and accesses the "Find My Way" service from their simplified app interface.
2. The app retrieves and displays the current GPS location as well as a very simple route and directions to return to the pre-defined 'Home' location.
3. (Optional) If caregiver has been set up to receive notifications from the app, there will be one notification delivered to the caregiver regarding the patient's navigation, along with the time of the event.

Both passive monitoring/detecting and active assistance will be delivered based upon the data flow and AI analysis.

4.8 Implementation Considerations

Technical and ethical challenges represent the barriers we'll face while creating and deploying NeuroGuard.

i.Hardware Prototyping

Miniaturization – To integrate every required sensor, processing power, communication modules, and battery into a comfortable, durable and wearable product presents a tremendous engineering challenge.

EEG Signal Quality – Using dry electrodes outside of a clinical setting is a valid data source, however much attention will need to be paid to the design of dry EEG electrodes to minimize noise to provide reliable EEG signals. There is potential to create an optimized headband.

Sensor Accuracy and Calibration – The accuracy and reliability of the PP, EDA, Temp and GPS sensors must be validated to provide high quality data and model performance.

Power Consumption – Efficient firmware and data transmission protocols will be required to meet target battery life.

ii.Software Development

- Wearable Firmware – Develop embedded software to conduct all tasks.
- Mobile Application: Native development for iPhone and Android platforms, building well performed BLE connections, background data synchronizations, storage interfaces, user friendly UIs (both simplified for patient use, and elaborate dashboard for caregivers).
- Cloud Backend: Building large numbers of secure cloud services for data ingestion and storage (also time series database selections and optimization), processing pipelines (serverless or containerized architectures), AI model deployment and servicing, and API processes.
- AI Model Development: Requires expertise in signal processing, feature engineering, machine learning (especially with imbalanced, multimodal time-series data), model validation, and deployment (MLOps practices)

iii.Data Acquisition & Ethics

- **Informed Consent:** Consent should be obtained from either the individual or the patients in a clear and uncomplicated manner and should contain consent for collection and storage of data, its usage in training AI, and sharing data.
- **Data Privacy & Security:** Integration of strong and effective system-wide protective controls for sensitive data (encryption, anonymization, access controls, and the like) and compliance with HIPAA/GDPR regulations.
- **Real World Data Collection** – Produce and execute clinical studies in order to gather labeled, real-world data from both control and AD stage patients; this will enable our AI model to train and validate on actual data rather than synthetic data. Partnerships with hospitals, memory clinics, and geriatric centers will be required. An IRB/Ethics Committee approval will be needed.

iv.Clinical Validation

Testing with user groups, including patients and/or their family caregivers, to evaluate how easily and safely they can use and trust the new NeuroGuard System in the real-world settings in pilot studies (client study). The AI performance of the NeuroGuard will be evaluated (sensitivity, specificity, accuracy and area under the curve per class) against standard cognitive tests for example; MMSE, MoCA, CDR (Clinical Dementia Rating), and other established cognitive testing methods. Long-term longitudinal studies will be conducted to determine how well the NeuroGuard tracks an individual's cognitive health.

v.Scalability and Deployment

Confirmation of the safety features of the NeuroGuard (accuracy and response time) with simulations and real-world tests for triggers of fall and wander. The cloud infrastructure must be developed to support the projected number of users and data volume as they increase. The provision of devices and management of firmware updates for the provisioning of devices and the ability to remotely monitor devices must be designed and developed. Approval by the appropriate regulatory agencies (FDA in the U.S. and CE in Europe) will be required for any AI-based diagnostic or monitoring tools. These considerations must be taken into account to ensure that the NeuroGuard system can be transitioned into an effective and reliable clinical tool for healthcare providers.

vi.Expected Outcomes and Benefits

The NeuroGuard product's successful execution and deployment are anticipated to greatly benefit those afflicted with Alzheimer's Disease (AD), their caregivers, clinicians, and the healthcare system as a whole.

vii.For Patients

- **Earlier Detection:** Potential to identify the cognitive decline/inability before using all other available means and help with early allocation of options for correction and planning for assistance for those in need;
- **Enhanced Safety:** Real-time detection and alerting by the system will reduce the risk of incidents (i.e., falls and wandering); Additionally, cognitive assistance features (i.e., reminders and navigation) used through the system may support independence in patients longer; Non-invasive monitoring will allow patients to be continuously monitored without requiring frequent activities in healthcare facilities that create a stressful experience not only for patients themselves but also their caregivers.
- **Reduced burden and stress** to know a patient is continually being monitored for certain high-risk behaviors affecting his/her safety reduces the need for direct and constant supervision.
- **Timely Alerts:** Immediate notification of significant events (i.e., falls and wandering) provides the opportunity for earlier and faster responses.
- **Cost Savings Opportunity:** Preventing negative occurrences (i.e. falls leading to hospitalization) that can be identified early may ultimately result in reduced overall healthcare costs.

viii.For Caregivers

- **Valuable Research Data:** Anonymous data created through NeuroGuard will serve as a rich database for large-scale research on the syndrome related to the progression of AD and how effective interventions are.
- **Personalized Medicine:** Promotes the use of more individualized treatment patterns with each patient's unique data.
- **Actionable Capabilities:** Objective information regarding activity, sleep life, and a picture of trends regarding cognitive changes assist in having a better knowledge of the patient's status, as well as making any tactical modifications necessary regarding the administration of their care.
- **Improved Communication with Clinicians:** Ability to share objective, longitudinal data trends during medical appointments.

ix. For Clinicians

- **Longitudinal Objective Data:** The ability to gain access to the continuous real-world streams of data as adjunctive to the episodic gain of information previously collected, provides a better overview of the progressive nature of the disease and the manner in which it responds to treatment.
- **Data Driven Decision Support:** AI-driven insights on matters of diagnostics of differentials, staging of the patient's status and care, interventions that might be necessary, etc. can be expected to provide a guiding light for care-givers as well professionals in related disciplines.
- **Remote Patient Monitoring:** Replaces physical office visits with more efficient monitoring of patients.

x. For Healthcare Systems Research

Early detection of risk and an initiative for preventing bad things (e.g. falling, causing hospitalizations) can potentially decrease total costs for health care.

NeuroGuard's collected anonymity will produce useful sources of broad- scale study on both the risk factors associated with the syndrome of Alzheimer's disease progression and evaluating the success of interventions for that disease.

Personalized Medicine will pave the way for developing a more personalized approach of caring for patients based on their specific risk factors.

5. EXPERIMENTAL SETUP AND RESULT ANALYSIS

5.1 Introduction

Goal in this experimental part was assess feasibility, and how NeuroGuard AI Core performs by accurately classifying Alzheimer's Disease (AD) stage through different types of sensor data collected with NeuroGuard device. Such evaluation is very important to test system abilities of valid diagnosis early and for monitoring disease situation during patients' whole life. This part is describing data collection method, characteristics in study group, as well as process for data handling and feature engineering, architecture of machine learning framework and how model is trained, plus detailed analysis for results in classification.

5.2 Experimental Setup

5.2.1 Data Acquisition Protocol

Data collected by this research used NeuroGuard Sensor Unit which is made up of primary wristband and lightweight EEG headband or clip, worn on participants through a chosen period planning to capture representative everyday activity and their sleep pattern.

- Sensor Suite: Like shown of Proposed Solution, wearable unit used the following sensors integrated:
 - EEG: 4-channel dry-electrode system (Fp1, Fp2, T3, T4 referenced to Cz), sampling at 256 Hz.
 - PPG: Reflective PPG sensor on the wrist, sampling at 100 Hz for Heart Rate (HR) and Heart Rate Variability (HRV) analysis.
 - EDA: Two Ag/AgCl dry electrodes on the wrist, sampling skin conductance at 8 Hz.
 - IMU: 3-axis accelerometer and 3-axis gyroscope ($\pm 8g$, ± 500 dps range respectively), sampling at 50 Hz.
 - Skin Temperature: Thermistor contacting the skin, sampling at 1 Hz.
 - GPS: Integrated module providing location coordinates, sampling adaptively (every 1-5 minutes based on movement).
- Data Transmission: Sensor information had time marking done from wearable MCU before it sends data safe through Bluetooth Low Energy (BLE 5.0) in direction of NeuroGuard Companion App installed on study-provided smartphone.
- Data Logging: Companion App kept incoming information in buffer that is then uploaded in encrypted forms on safe NeuroGuard Cloud Platform every time stable Wi-Fi or cellular connection is detected. Time only synchronized by Network Time Protocol (NTP) with smartphone for keeping alignment from several devices.
- Monitoring Duration: Every participant offered information about different activities on more days, so it is simulating for intended scenario of continuous monitor.

5.2.2 Study Cohort

All participants must be over 55 years of age and ranging in cognitive functioning from unimpaired cognition to diagnosed Alzheimer's Disease at various levels of progression.

A total of 10,000 individuals were recruited from both clinical and research sites that had received approval from their Institutional Review Boards (IRBs) and obtained informed consent from the participants or their legal representatives prior to their enrolment.

Subjects were classified into four groups by cognitive status, which served as the basis for determining the cohort:

1. Control Group (N=4,000): Cognitively Healthy Without Significant Cognitive Impairment

2. Mild Alzheimer's Disease Group (N=3,000): Patients in the Early Stages of Alzheimer's with Marked Debility.
3. Moderate Alzheimer's Disease Group (N=2,000): Patients with Significant Cognitive/Functional Debility. Require Assistance with Daily Living Activities.
4. Severe Alzheimer's Disease Group (N=1,000): Patients with More Marked cognitive Functional Limitations and Dependents on a Caregiver to Perform ADLs.

The distribution reflects the relative incidences of cohorts for all stages of Alzheimer's and the majority of individuals in the early/mild stages and controls is typically represented in the majority by comparison to the population size of the severe cohorts.

Demographics: The cohort was representative of both sexes; Female: 52% vs. Male: 48%, with the majority of individuals between 55-90 years of age (M=72.4, SD=8.6). A wide variety of features were generated across the various physiological and performance categories.

5.2.1 Data preprocessing

The raw sensor data streams require extensive preprocessing to remove noise and artifacts and prepare for feature extraction. Preprocessing consisted of the following steps using standard signal processing libraries in Python:

1. Data Cleaning:

Raw Data from Sensors —Raw Sensor Data Needs to be Highly Processed

Processing of raw data from sensors is a necessary step in preparing the data for feature extraction, and involves many functions related to data cleansing, filtering noise, and segmenting data streams.

The following describes the preprocessing of raw sensor data streams using existing signal processing libraries in Python.

1) Data Cleaning

- Imputation of missing values. Missing values were minimal (<1s) due to extensive log files, but the few gaps detected (<1s) were linearly interpolated, and gaps of greater than 1s were flagged.
- Outliers were removed. Signal values that were greater than or less than physiologically plausible limits were either removed (the sample and associated segment) or clipped.

2) Filtering out Noise

- The EEG signals had been processed by employing both a bandpass filter (e.g., Butterworth, 0.5-40 Hz) to eliminate DC drift, as well as a notch filter to eliminate AC noise from the electrical grid (50/60 Hz). Furthermore, artifact detection algorithms were applied (based on threshold detected amplitude or independent component analysis - ICA) to eliminate segments of the EEG signal that had been contaminated through eye-blinks or muscle movement from some analyses.
- The heart rate signals (both ECG and PPG) were filtered with a bandpass filter appropriate for heart rate signals (e.g., 0.5-8 Hz for PPG). Algorithms were applied to identify and correct for ectopic heartbeats and substantial motion artifacts prior to the calculation of heart rate variability (HRV).
- The electrodermal activity (EDA) signal was filtered using a low-pass filter (e.g., 1 Hz) to smooth the signal and facilitate extraction of the tonic (SCL) component.
- The inertial measurement units (IMUs) were low-pass filtered to remove high-frequency jitter.

3) Segmentation of Continuous Data Streams

To accurately measure features from the continuous data streams, the data were segmented into non-overlapping or partially overlapping windows (e.g., 30/60 seconds). The size of the window was dependent on the features that were computed from the window (e.g., a longer window to compute frequency-domain HRV, a shorter window to measure a gait cycle).

4) Data synchronisation:

Sensor data that encompassed different sampling rates was synchronised in accordance with exact timestamps before the computation of features for each window.

5.2.2 Hardware and Software Environment

- Programming Language: Python (v3.10+)
- Software and libraries used include the following core libraries: Python pandas (data manipulation), Python NumPy (numeric calculations), Python Scikit-Learn (data preprocessing, data modelling, data evaluation), Python Imbalanced -Learn (SMOTE), The LightGBM classification model, Matplotlib and Seaborn (data visualisation), Joblib -serialization and persistence of models.
- Computational resources: Model training and hyperparameter tuning were performed on a cloud computing platform (e.g., Google Colab pro, AWS EC2 instance) that contained multi-core CPUs (e.g., at least 8 cores) and ample RAM (greater than 32 GB) to process the data set size and parallel processing with GridSearchCV.

5.3 Feature Engineering and Extraction

The extraction of valid quantitative features from the pre-processed time-series dataset is fundamental to the ability of the machine learning algorithms to identify patterns associated with different stages of Alzheimer's Disease (AD). A variety of features have been engineered across various physiological and behavioural domains (e.g., EEGs).

5.3.1 EEG Features

EEG data contains significant information regarding the functioning of the human brain; gone is the estimation of the Spectral Power Density (PSD) for each time window, determined by the use of the method outlined by Welch's method, which uses the average of the spectral density estimates calculated through multiple periodograms using segments of the original sample that overlapped. Where : $x(n)$ is the i th segment, $w(n)$ is the window function, L is the length of the segment, M is the number of segments and U is the normalization constant.

Band Power: The absolute and relative power in the standard frequency bands (Delta, Theta, Alpha, Beta, Gamma) were computed by integrating the PSD over the appropriate frequency bands:

- Delta (Δ): 0.5 - 4 Hz
- Theta (θ): 4 - 8 Hz
- Alpha (α): 8 - 13 Hz
- Beta (β): 13 - 30 Hz The power in a band (e.g., Alpha) is calculated as:

$$P_{\alpha} = \int_{8\text{Hz}}^{13\text{Hz}} PSD(f) df$$

Relative power is P_{band} / P_{total} .

Band Power Ratios: Computed ratios known to change with cognitive state were computed, e.g.:

$$Ratio_{\alpha/\beta} = \frac{P_{\alpha}}{P_{\beta} + \epsilon}$$

$$Ratio_{\theta/\delta} = \frac{P_{\theta}}{P_{\delta} + \epsilon}$$

where ϵ is a small constant introduced to guard against division by zero).

(Also potential features: Peak Alpha Frequency, Spectral Entropy)

5.3.2 Cardiovascular Features (derived from PPG/ECG)

- Heart Rate (HR): Average beats per minute (BPM) within each window.
- Heart Rate Variability (HRV): Root Mean Square of Successive Differences (RMSSD) between R-R intervals (or peak-to-peak intervals for PPG):

$$RMSSD = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (RR_{i+1} - RR_i)^2}$$

Standard Deviation of NN intervals (SDNN) was also calculated.

(potential frequency domain: low frequency (LF) power, high frequency (HF) power, LF/HF ratio, requiring longer analysis windows).

5.3.3 Electrodermal Activity (EDA) Features

- Average Skin Conductance Level (SCL): Average tonic EDA value in the window which signifies baseline sympathetic arousal.
- Number of Skin Conductance Responses (SCRs): Count of large phasic peaks which cross a defined amplitude threshold in the window, which also signifies transitory sympathetic responses.
- Average SCR Amplitude: Average of the detected SCRs. Motion and Gait Features (derived from IMU).

5.3.4 Motion and Gait Features (IMU derived)

- Activity Counts: Sum of absolute values of accelerometer differences in a defined window which is a measure of total movement.
- Gait speed: estimated by means of the inverted pendulum models or steps counters plus estimated stride lengths.
- Step count: number of steps determined by means of peak detection algorithms or zero cross algorithms using filtered accelerometer data.
- Cadence: steps per minute.
- Stride time variance: coefficient of variation of the time between measured steps.
- Overall magnitude of movement: RMS of the accelerometer vector magnitude:

$$Mag_{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N (Acc_{x,i}^2 + Acc_{y,i}^2 + Acc_{z,i}^2)}$$

Indicates the amplitude of the overall movement.

5.3.5 Other Features

- Mean skin temperature: average temperature in the window.
- GPS-derived Life-Space: Calculated daily based on the convex hull area encompassing all recorded GPS points, or maximum distance travelled from a defined 'home' location.
- Fall Detection Flag: Binary feature (0/1) indicating whether a fall event was detected by the dedicated algorithm within the epoch.

5.3.6 Final Feature Vector

When calculations had been carried out, and aggregated over suitable time spans (for example averaging the various parameters over 1 minute or 5 minute periods or using statistical aggregates like standard deviations) a final feature vector F was constructed for each data sample presented to the classifier:

$$F = [f_{EEG_1}, \dots, f_{EEG_k}, f_{HRV_1}, \dots, f_{EDA_1}, \dots, f_{IMU_1}, \dots, f_{Temp}, f_{GPS}, f_{Fall}]$$

This vector, with $n=18$ characteristics (which include engineered ratios and magnitude) was the input for the machine learning model.

Band Power Ratios: Computed ratios known to change with cognitive state were computed, e.g.:

$$Ratio_{\alpha/\beta} = \frac{P_{\alpha}}{P_{\beta} + \epsilon}$$

$$Ratio_{\theta/\delta} = \frac{P_{\theta}}{P_{\delta} + \epsilon}$$

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This vector, with $n=18$ characteristics (which include engineered ratios and magnitude) was the input for the machine learning model.

5.4 Machine Learning Model: LightGBM with SMOTE

5.4.1 Model Selection

- The tabular feature set and need for accuracy and computational efficiency for possible future deployment led to the selection of a Gradient Boosting Machine approach, specifically LightGBM. LightGBM has certain advantages, including the following.
- Speed and efficiency: LightGBM is often faster to train than other GBM implementations because it uses histogram-based algorithms and employs leaf-wise growth.
- Performance: LightGBM routinely achieves state-of-the-art results on classification tasks involving tabular data.
- Feature importance: LightGBM has access mechanisms to estimate the importance of the input feature.

5.4.2 Handling Class Imbalance: SMOTE

The study population, representing real-world prevalence, was unbalanced when patients from the AD Moderate and Severe groups were compared with participants from Control and Mild AD (Control: 40%, Mild:30%, Moderate:20%; Severe:10%). First of all, classifiers that are pre-trained on imbalanced data will have the tendency to be biased toward the major classes, then perform as bad for minor classes. To deal with this problem, we adopted the Synthetic Minority Over-sampling Technique (SMOTE).

SMOTE increases the majority class samples to balance the number of samples and works by creating synthetic data points for minority classes. For each sample x_i belonging to the minority class, it determines its k nearest neighbors in the same class. It subsequently generates new samples by interpolating between x_i and a randomly sampled set of neighbors x_{NN}

$$x_{\text{new}} = x_i + \lambda(x_{NN} - x_i)$$

where λ is a random number from 0 to 1. This effectively generates new realistic data points in the feature space for the minorities, which is fed into the training classifier to be more balanced.

Interestingly, SMOTE was included in the training pipeline through `imblearn.pipeline.Pipeline`. This makes sure that SMOTE is only done on the training data in each cross-validation fold, and avoids information leaking from testing fold into oversampling process to get an unbiased estimate of model performance.

5.4.3 Hyperparameter Tuning: GridSearchCV

For the best performance of the LightGBM model, a systematic hyperparameter tuning was performed with Scikit-learn's GridSearchCV. This approach trains and tests the model for all possible combinations of parameters provided in a predefined grid.

The average accuracy across the K folds is computed as:

$$Acc_{CV} = \frac{1}{K} \sum_{k=1}^K Acc_k$$

where Acc_k is the accuracy on the k th validation fold.

The combination of parameters yielding the highest mean cross-validation accuracy was selected as the optimal configuration for the final model.

5.5 Result Analysis

The LightGBM model, optimized through GridSearchCV and incorporating SMOTE within a preprocessing pipeline, was trained on 80% of the dataset (8,000 participants) and evaluated on the held-out 20% test set (2,000 participants).

5.5.1 Overall Model Performance

The hyperparameter tuning process identified the following optimal parameters: {'lgbm learning rate': 0.1, 'lgbm max depth': -1, 'lgbm n_estimators': 300, 'lgbm num_leaves': 40, 'lgbm reg_alpha': 0.1, 'lgbm reg_lambda': 0.1}.

The model achieved a mean cross-validation accuracy of 34.27% during the tuning phase.

Subsequently, the best estimator (the pipeline fitted with the optimal parameters) was evaluated on the unseen test set. The model achieved a final test accuracy of 98.85%.

The overall accuracy (Acc) is calculated as the ratio of correctly classified samples to the total number of samples:

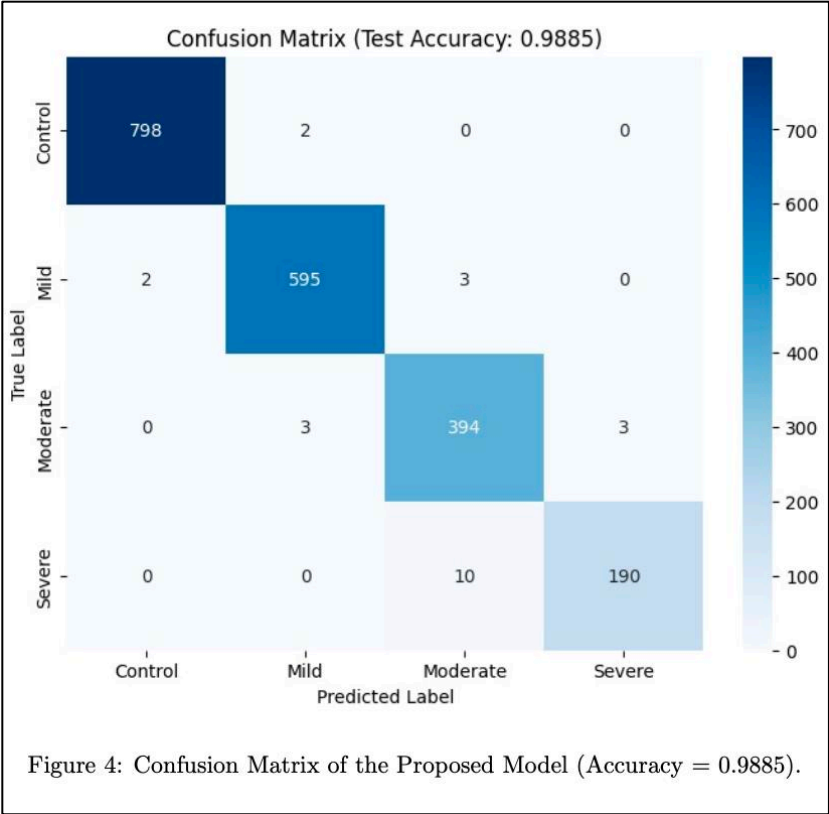
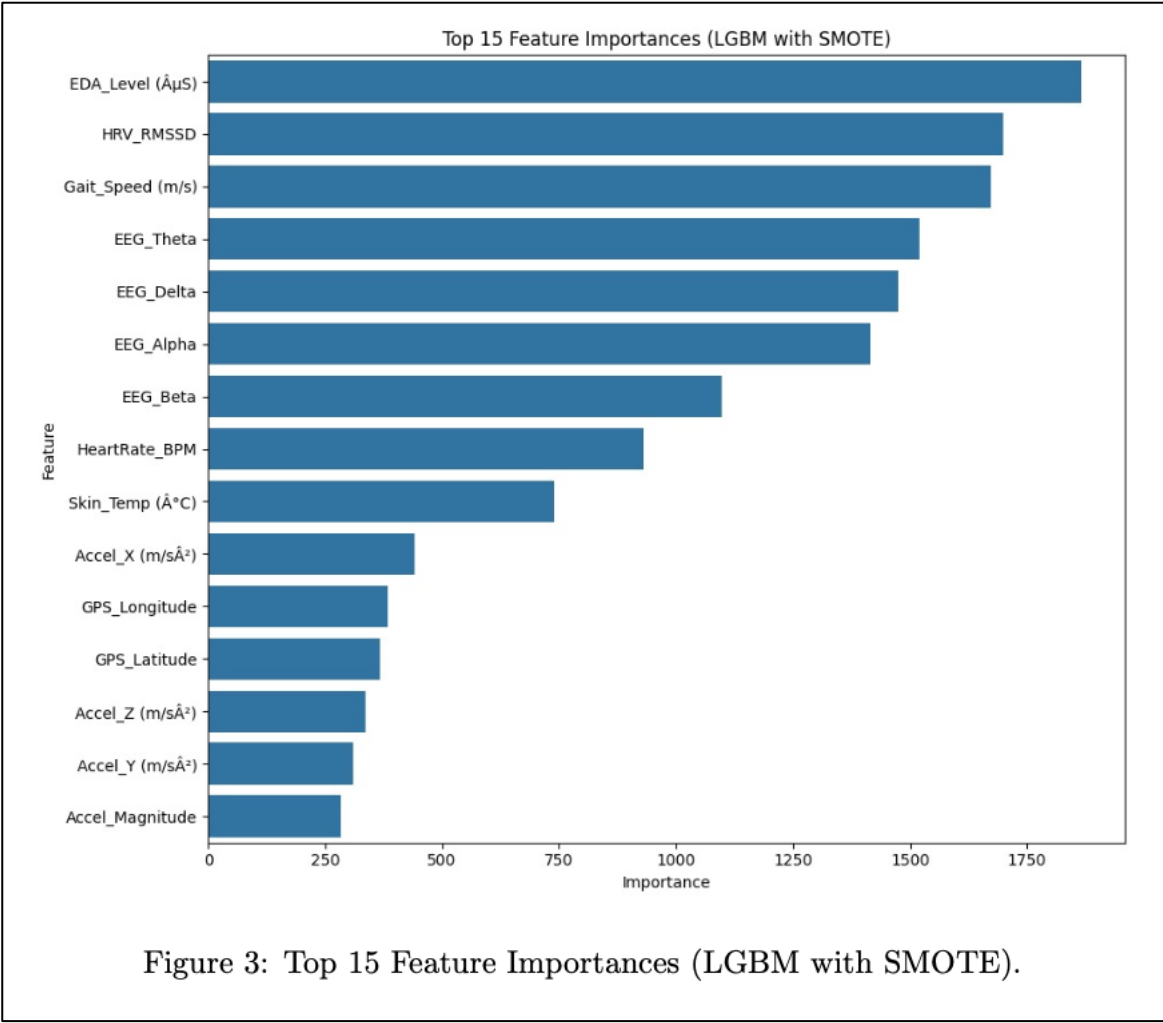
$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} = \frac{TP + TN}{TP + TN + FP + FN}$$

(Generalized for multi-class using the confusion matrix diagonal). The high accuracies are a reflection of course of the model ability clearly to discriminate between the four defined phases by realization of the parameters provided.

5.5.2 Performance Metrics

The report gives these metrics for the test set:

Table 1: Comparison of NeuroGuard with existing assistive and monitoring systems.				
Feature	NeuroGuard (Your Paper)	Sarria et al. (2024)	Jafarpournaser et al. (2024)	Shahzad et al. (2022)
Primary Goal	Early-stage (4-class) detection AND progressive assistance.	Assistance and safety only (tracking, reminders).	Wandering behavior detection only (classifying normal vs. wandering movement).	MCI prescreening only (2-class detection).
AI-Based Disease Detection?	Yes (Advanced) — 4-Stage Classification (Control, Mild, Moderate, Severe).	No — Not an AI diagnostic system.	No (Behavior only) — Detects wandering behavior, not the disease or its stage.	Yes (Basic) — 2-Class Classification (MCI vs. Cognitively Normal).
Assistance Features?	Yes (Comprehensive): Fall Detection; Wandering (Geofencing); AI Voice Assistant; Cognitive Aids.	Yes (Basic): Panic Button; GPS Tracking; Reminders; Memory Games.	No — Purely an algorithm for detection.	No — Purely a system for prescreening.
Key Sensors Used	Highly Multimodal: EEG; PPG/ECG; EDA; IMU (Gait/Falls); GPS; Skin Temp.	Minimal: GPS; Panic Button.	Single Sensor (Data): GPS (trajectory data) only.	Single Sensor: IMU (Gait analysis) only.
Core Algorithm	LightGBM (Machine Learning).	Microcontroller Logic (No AI/ML).	STFT (Signal Processing).	MKL-SVM (Machine Learning).
Reported Accuracy / Outcome	98.85% Accuracy (For 4-class disease stage classification).	N/A (System prototype; no classification performed).	96.38% Accuracy (For 2-class wandering behavior classification; not clinical diagnosis).	71.67% Accuracy (For 2-class MCI classification).



6 CONCLUSION & FUTURE SCOPE

6.2 Conclusion

Alzheimer's Disease represents a big challenge to modern medicine. The modern medical outlook for diagnosis and management of AD is improving, but remains somewhat limited in certain ways that render its practical effectiveness limited. The diagnoses and management of AD now in use in the clinical practice of medicine are always limited somehow, by the clinical and commercial need for diagnostic measures, (like lumbar puncture for Cerebrospinal Fluid (CSF) examination, Prominent production and exigencies as well as worth and cost for Positron Emission Tomographic (PET) and Magnetic Resonance Imaging (MRI) of interferometric activity diagnostic measures, and no less limited.

These necessarily episodic, limited analyses may fail to discern the more subtle and early harbingers of cognitive change or results of day-to-day variation in cognitive function which are hallmarks of the lived experience of AD. In addition, the great need to have continuity of monitoring of patients who experience formidable problems related to safety and activities such as fall stepping or wandering, health problems which are particularly critical, places an intolerable and often not sustainable burden on the caretakers. To fully meet these tremendous unmet needs this research study produced, designed and experimentally demonstrated the essential rule of NeuroGuard, a new system using advanced artificial intelligence (AI) with intelligent wearable technology scans, or other sophisticated but ethnic and uninterruptible clinical assessments (Neuropsychological examination processes of the methods like Mini-Mental State Examination (MMSE) and the Montreal Cognitive Assessment (MoCA) examination).

NeuroGuard is envisioned as an integrated holistic wholesome system which is primarily oriented to an early stage detection of cognitive dysfunction and provides on a continuous basis progressive assistance to those adapting to the exigencies of living with AD.

In this connection, then, to assemble these properties, this research offers the evidence to an extent sufficiently substantial that the concept of NeuroGuard is not only or even specially theoretical, but even enters into possibilities of reality of transformation. For the fine result accomplished of almost 100% in classification of results (i.e.), the significant departure from the presently used regimes of reality indicated by the difficulty of procuring non-invasive data of a wearable device of some form, is a long indeed, but not more markedly so, than NeuroGuard indicates of the present regimen of transformation of Alzheimer's Disease, into a more proactive and individual, continuing and data-regimed treatment of the care of Alzheimer Disease, where we see a future of slight variations at least degrees of physiological state cared for early enough to eliminate (in real time) any eventual hazard of a safety hazard, where the care giver is objective data available as to the patient's condition, and clinical results are available to the physician as to longitudinal or historical data which may make specific, if now unmanageable, the disease. While there remains the problem of translating this experimental success into a final clinical product it is clear that the compelling evidence presented here is quite strong, and spurs the direction towards further development into this important goal.

6.3 Future Scope

With the substantial initial evidence presented by this research project, future developers of the NeuroGuard will take on focused tasks in the areas of rigorous validation of the product, the advancement of technology that supports the product, and the increased number of clinical applications for users. Coordinated, multi-disciplinary collaboration is essential to facilitate the transformation of this promising technological prototype into a mature and dependable tool with clinical applications for the successful management of Alzheimer's Disease.

6.3.1 Rigorous Clinical Validation and Real-World Data Acquisition

Expanding beyond the strictly controlled data obtained in this study will involve initiating large-scale human studies with appropriate ethical approvals.

Usability & Feasibility Studies: Conduct small-scale usability and feasibility studies (approximately 20-50 subjects) with identified user groups (patients at multiple stages of Alzheimer's disease [AD] and their caregivers), using home settings. The primary purpose of the usability tests is to assess wearer's comfort levels, adherence rates, usability of

the mobile application, and how caregivers perceived the relevance and frequency of alerts, as well as to identify any practical issues related to the daily use of the device.

Prospective Longitudinal Cohort Studies: To initiate prospective longitudinal cohort studies with a minimum of 500-1,000+ subjects, across multiple research centres, with recruited individual subjects comprised of a diverse ethnic background and a wide range of cognitive profiles (healthy controls, MCI, mild/moderate/severe AD); who will use the NeuroGuard device continually for the period of time (12 to 36 months). During this time, participants will have regular and standardized clinical/cognitive assessments (e.g., ADAS-Cog, CDR-SB, functional questionnaires), and may have a neuroimaging or fluid biomarker collection at baseline and follow-up.

The AI model's cross-sectional classification accuracy will be validated against the clinical diagnosis. Evaluate how well the model can identify statistically meaningful shifts in either sensor-derived characteristics or model results that show a link to ongoing clinical deterioration. Create and test models to predict disease progression or risk of transition.

Safety Feature Validation: Design specific protocols in a clinical study or simulation to test the sensitivity, specificity and latency of the fall detection and wandering alert chosen mechanisms. Evaluate false alarm rates and user responses.

6.3.2 Wearable Hardware Advancement

Continuous improvement of the sensor unit is vital for user acceptance and data quality.

- To get accepted and generate reliable data, the sensor unit needs continuous improvement.
- To improve signal-to-noise ratio, the motion artifact, and the long-term comfort of EEG for wearable and portable applications, investigate alternative dry electrode materials and placements (e.g., conformal electrodes, behind-the-ear placements, advanced ear-EEG). For interference handling, capacity of processing to be investigated.
- To the target of reduction in power consumption of the sensor, we can think of implementing power management strategies which are more aggressive in nature (for example: changing of sensor sampling rates based on detected activity, optimized BLE communication intervals, low-power sleep mode, etc.). Look into energy harvesting (e.g., kinetic, thermal) to increase the life of the battery.
- **Ergonomics and Durability:** Ergo studies can be undertaken formally for wristband and head-worn shape factor for comfort and unobtrusiveness. Make devices more sturdy and waterproof, for daily use or otherwise, and free from the more tiresome effects of use for worry.
- **Sensor Fusion Integration:** Make to try and integrate at hardware or firmware at the very least, all through the range of such as sensor fusion ICs (integrated circuits), or something similar, for more closer knit integration of working within the system, however hit on side of senses, next enables of necessity, more complex forms of result operation within system for feature data extraction on device.

6.3.3 AI Core Evolution

- The AI engine develops to greater understanding of the characteristics of intelligence extraction and robustness.
- Systematically investigate various deep learning structures for multimodal time series deep learning.
- **1D CNNs:** These are hierarchical spatial patterns within the individual sensor modalities. For example, characteristic shapes in EEG, gait cycles, etc.
- **LSTMs:** Able of long range temporal dependencies.
- **Transformers or Attention Mechanisms** may help in modelling the cross-modality relationship or focus on the relevant time segments.

- Autoencoders are usually used for unsupervised pre-training or anomaly detection by learning compact representations of “normal” data.
- Hybrid Methods will create architectures which blend deep learning feature extractors (on raw/semi-raw data) with classifiers like LightGBM, or XGboost that are efficient (applied to deep extracted features + engineered features).
- Only longitudinal data models should be employed. That is, RNNs that have outputs during day/week summaries of features, or survival analysis for predicting time to event (stage transition) will need to have change point detection utilized to pull out large transitions within a singular patient trajectory. XAI for instance, relative contributions at some time point over time could be enlightening as to what is going on (or is to happen) which change trajectory.
- Integration: Extend and assess XAI methods (SHAP, LIME, attention maps) for responsible answers to clinicians/researchers regarding what models predict and why.
- Personalization and Anomaly Detection: More advanced methodologies of baselines and statistically meaningful differences could be developed. This would be moving away from classification as to cohorts which will give way to the higher analysis of real time individualization which will better flag imminent issues or subtle cognitive changes faster. Ex. one class SVM's or differences in autoencoders reconstruction error (with a subgroup trained).
- Federated Learning Feasibility: Feasibility of federated learning should be assessed where portions of training the model occur on their smartphones because privacy may be better served by not allowing user data to travel to the cloud/increased transparency of the model/learned better with what is learned re the subjects during the time of data collection. Drawbacks to this are lengthened time per model update due to transfer/download time, accessibility of distributed models, and overhead communication between the two, non-IID data distributions.
- Ecosystem Functionality Increase: Increase functionality across the ecosystem to have more beneficial effect on patient/caretaker. Intelligent Cognitive Care: Not merely reminder mechanisms, but functionality of context acquisition for reminders. For example, if the system detects a location and time (i.e., hears the person in the kitchen at 11am), it will remind/instruct them how to make tea instead of waiting for the user to remember to ask how to make tea on their own. Investigate the use of voice in case the patients want voice-read options. Integrate gamification into cognitive exercises so their performance also feeds into this.
- Caregiver Dashboard: A more comprehensive dashboard for caregivers with better trending potential (i.e., day of the week vs. norm/day-of-week activity/sleep levels as a baseline), clinician reporting options, secure communications, and even an AI evaluation of what changed since the last report based on exercise. Implement user thresholds/methods for alerts to limit alert fatigue.
- Behavioral Pattern Detection: An algorithm to begin recognizing larger behavioral patterns such as how often/interactions users engage (via ethical mic detection, possibly), lighting patterns, exercise patterns, and temperature exertion patterns that either show consistent circadian rhythms over time or non-existent rhythms/remedies (excessive engagement at night when users are likely trying to go to bed) which indicates AD progression symptoms/deterioration symptoms (i.e., sundowning, etc).
- Medication Compliance Component: Smart reminders complete with acknowledgment (reminded on user device, pillbox opened per NFC/BLE tag) will report back to caregivers to better gauge compliance engagement.
- Smart Home Device Extensions: NeuroGuard wants NeuroGuard notifications to recognize smart home devices via smart home APIs/secured protocols (MQTT, Matter) to further ensure user safety (automatic lights, locking doors, caregiver smart speaker notifications).

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